Lab 2 — Agile Backlog & Sprint Simulation

Customizable Subscription Box Scheduler · PES University, Dept. of CSE · Section C

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Jira project Customizable Subscription Box Scheduler (key CSBS), Scrum, company-managed. Backlog derived from the Lab 1 requirements table (five functional requirements, two non-functional requirements).

1. Epics and User Stories

Three Epics covering the five functional requirements and two non-functional requirements from Lab 1. Seven core User Stories, 37 story points, estimated on the Fibonacci scale, all scheduled into Sprint 1.

Epic 1: Subscriber Box Customization

Traces to FR-001, FR-002, FR-003

Lets an authenticated subscriber swap or pause a box, filter products by preference tags, and skip a single billing cycle, without ever having to cancel the subscription.

Story	User story	Priority	Points	Sprint
1.1 Swap or Pause Monthly Box	As a subscriber, I want to swap items or pause my monthly box at least 48 hours before renewal, so that I can control my box contents and avoid unwanted deliveries.	High	5	Sprint 1
1.2 Filter Products by Preferences	As a subscriber, I want to select preference tags such as coffee, fiction, or vegan, so that I can see products matching my interests.	High	3	Sprint 1
1.3 Skip Upcoming Billing Cycle	As a subscriber, I want to skip one upcoming billing cycle without cancelling my subscription, so that I can temporarily stop receiving a box while keeping my subscription.	High	3	Sprint 1

3 stories, 11 story points.

Epic 2: Fulfillment Management

Traces to FR-004

Gives the Fulfillment Lead a single, timely source of truth for what has to be picked, packed, and shipped each billing cycle.

Story	User story	Priority	Points	Sprint
2.1 Generate Fulfillment Manifest	As a Fulfillment Lead, I want to receive a manifest containing all confirmed box selections for the current billing cycle, so that I can accurately pick,	Medium	8	Sprint 1

	pack, and ship subscriber boxes.			

1 story, 8 story points.

Epic 3: Delivery Tracking and Security

Traces to FR-005, NFR-001, NFR-002

Covers the subscriber-facing delivery-status view together with the two non-functional constraints — bulk label export speed and encryption of subscriber data — that keep the operation trustworthy at scale.

Story	User story	Priority	Points	Sprint
3.1 View Delivery Status	As a subscriber, I want to view the delivery status of my current box, so that I know whether my box is processing, shipped, or delivered.	Medium	5	Sprint 1
3.2 Fast Shipping Label Export	As a Fulfillment Lead, I want shipping labels for 10,000 monthly boxes to be exported in under one minute, so that large fulfillment operations are not delayed.	High	8	Sprint 1
3.3 Secure Subscriber Data	As a subscriber, I want my payment and personal information to be securely encrypted, so that my sensitive data is protected.	High	5	Sprint 1

3 stories, 18 story points.

Process validation story (outside Lab 1 scope)

A further single-story sprint was run purely to satisfy the lab's two-sprint requirement and to validate the Sprint 1 release, rather than to trace to a new functional requirement.

Story	User story	Priority	Points	Sprint
S2.1 Sprint 2 Validation and Review	As the project team, we want to validate the completed subscription box scheduling features and review sprint performance, so that Sprint 1's output is confirmed before closing the lab exercise.	Medium	3	Sprint 2

1 story, 3 story points.

2. Sprint results

Sprint	Started	Items	Points	Completed
CSBS Sprint 1	08/Sep/26 (approx.)	7	37	7 of 7
CSBS Sprint 2	08/Sep/26 10:45 AM	1	3	1 of 1
Remaining backlog	—	0	0	—

> **Sprint 1 goal.** A subscriber can customize, skip, and track their monthly box, and the fulfillment and security operations behind it are fully in place.

> **Sprint 2 goal.** Validate the completed subscription box scheduling features and review sprint performance before closing the exercise.

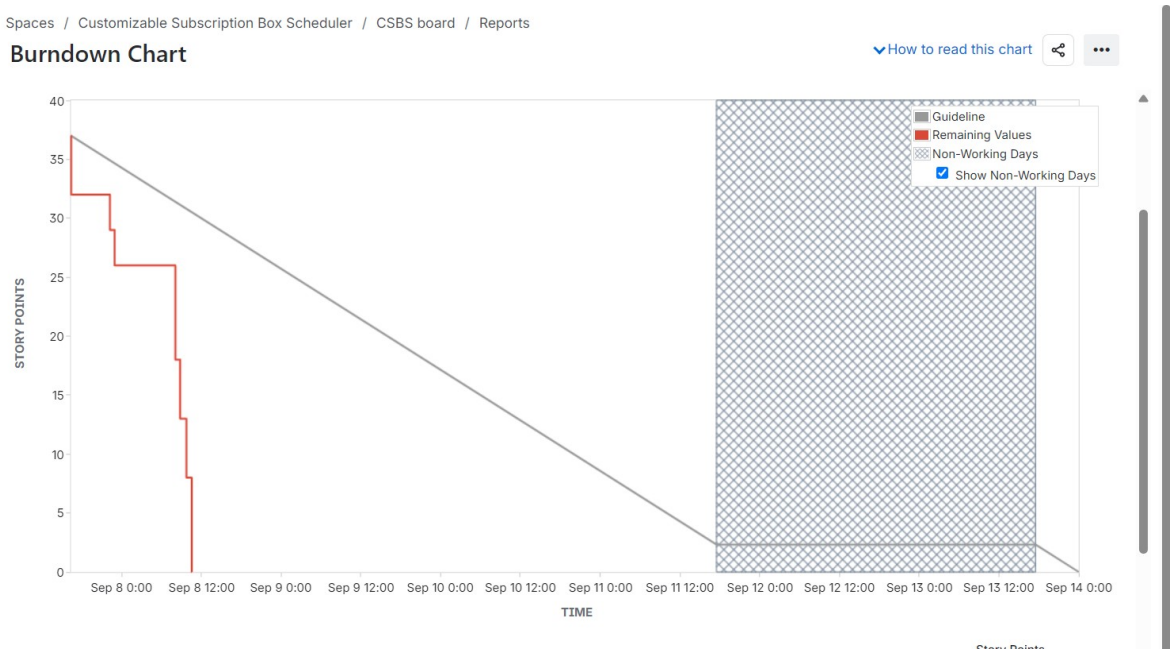
Both sprints were time-boxed to one week. Sprint 1 closed completely (7 of 7 stories, 37 of 37 points) and Sprint 2 closed completely (1 of 1 story, 3 of 3 points). No backlog items remain outstanding; Sprint 3 was auto-created empty and was not started.

3. Burndown charts

In both charts the Remaining Values line falls from the committed total to zero within the first half-day of the sprint and then runs flat along the zero axis for the rest of the one-week time box, while the guideline descends slowly across the full week. The line sits well below the guideline not because subscriber-facing work genuinely took only a few hours, but because both sprints were executed as a classroom simulation — the whole backlog was dragged through To Do, In Progress, and Done in a single sitting rather than over five real working days. Read as an actual team's velocity, this shape would be the signature of severe under-commitment; read as a simulation exercise, it simply reflects that the lab compresses a week of Scrum ceremony into one continuous session.

CSBS Sprint 1 — 37 story points

Jira Burndown Chart for Sprint 1: remaining Story Points falling from 37 to 0.



CSBS Sprint 2 — 3 story points

Jira Burndown Chart for Sprint 2: the 3 Story Points reduced to 0.

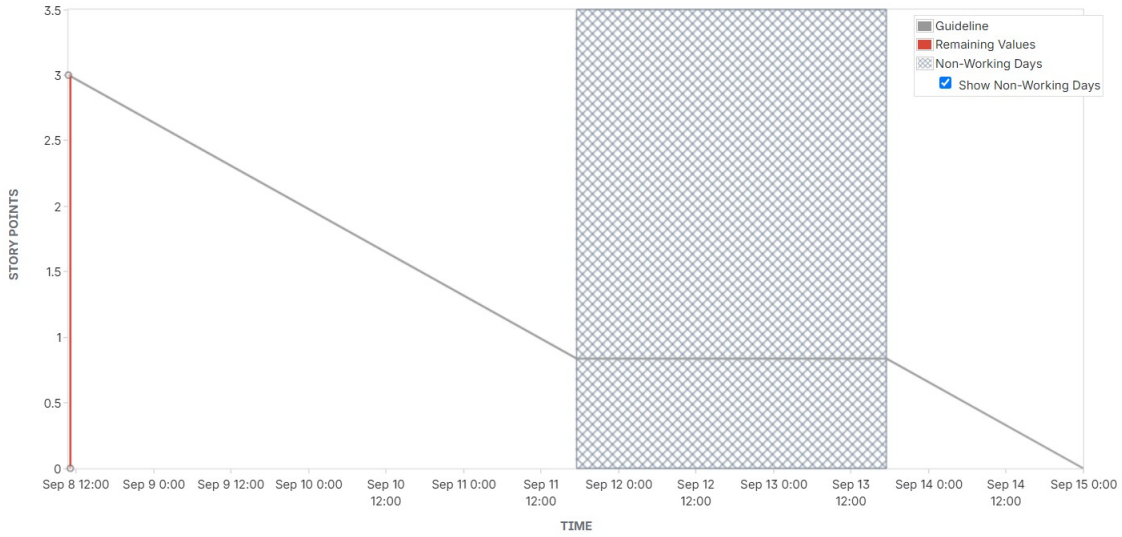
Spaces / Customizable Subscription Box Scheduler / CSBS board / Reports

Burndown Chart

How to read this chart

CSBS Sprint 2 | Story Points

Validate completed subscription box scheduling features and review sprint performance.

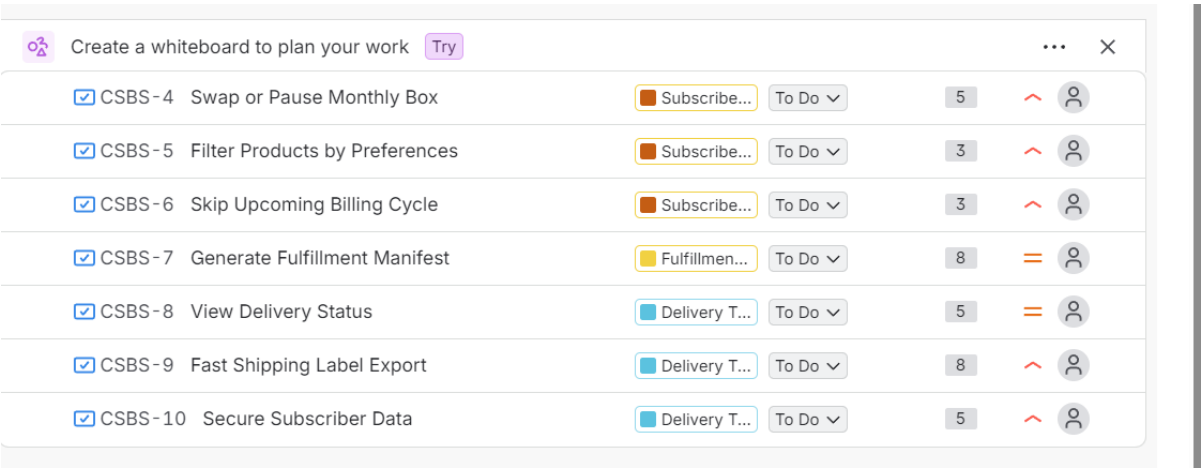


Date	Issue	Event Type	Event Detail	Story Points Inc.	Dec.	Remaining
08/Sep/26 10:45 AM	CSBS-11	Sprint start		3		

4. Jira workspace

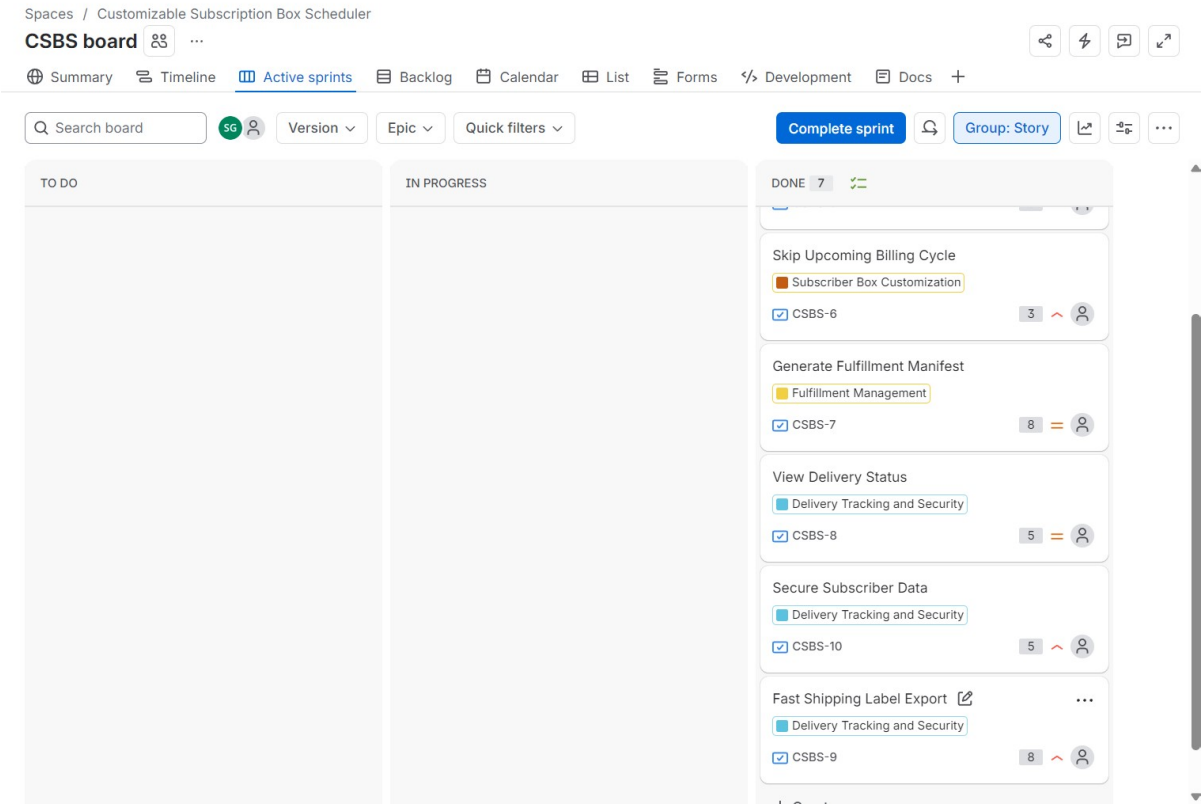
Backlog with Epics and User Stories

Backlog panel showing all seven core User Stories grouped under their Epics, with priority and Fibonacci story points visible on each row.



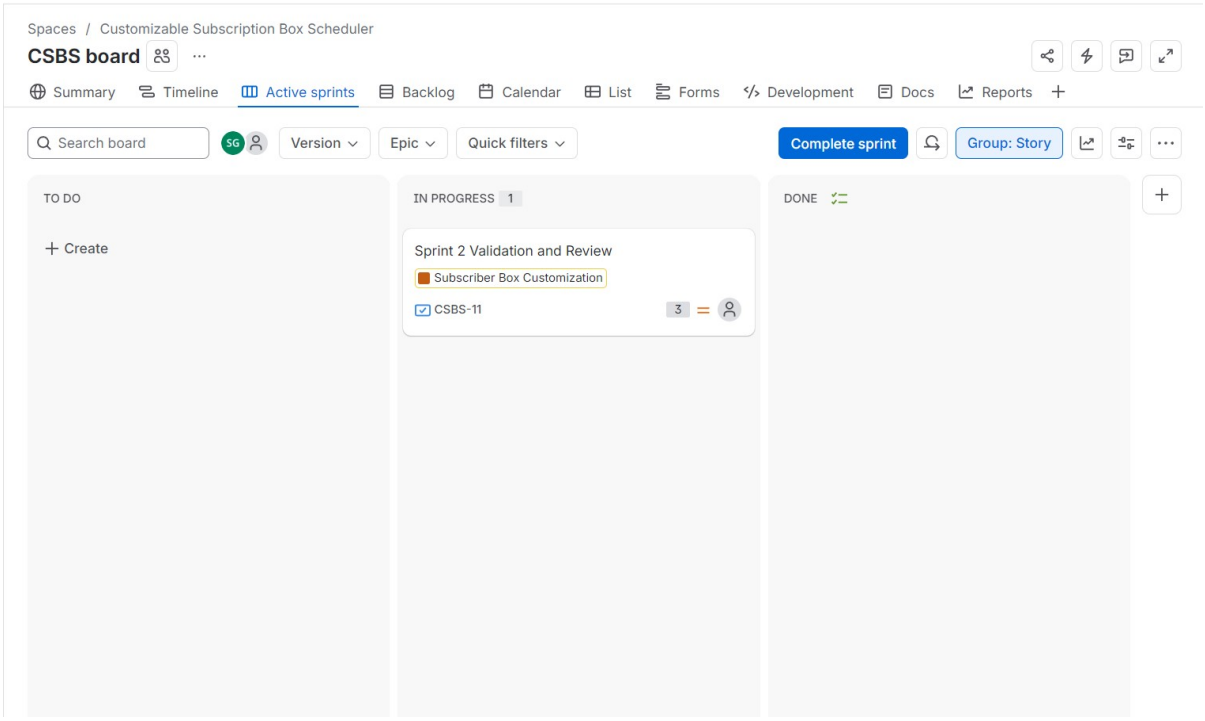
Sprint board — CSBS Sprint 1 (completed)

Active sprint view at completion: 0 To Do, 0 In Progress, 7 Done.



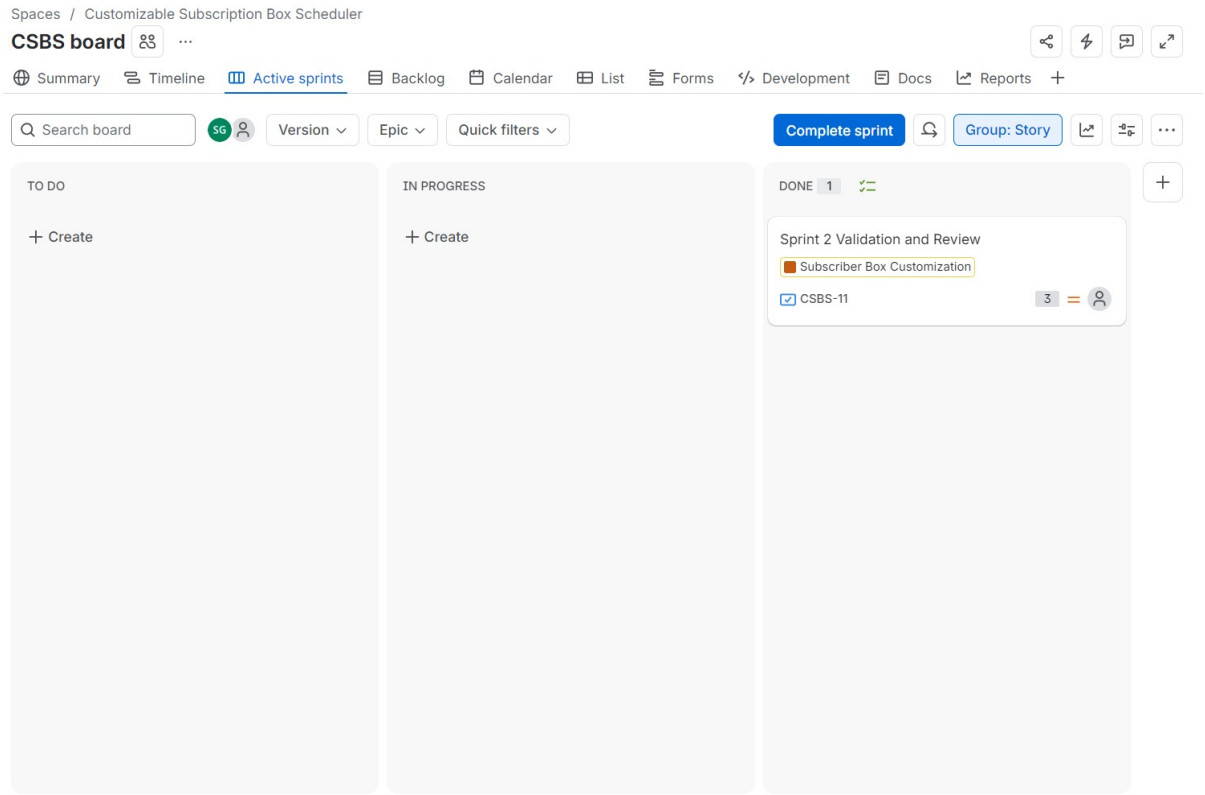
Sprint board — CSBS Sprint 2 (mid-sprint)

Active sprint view with CSBS-11 moved into In Progress: 0 To Do, 1 In Progress, 0 Done.



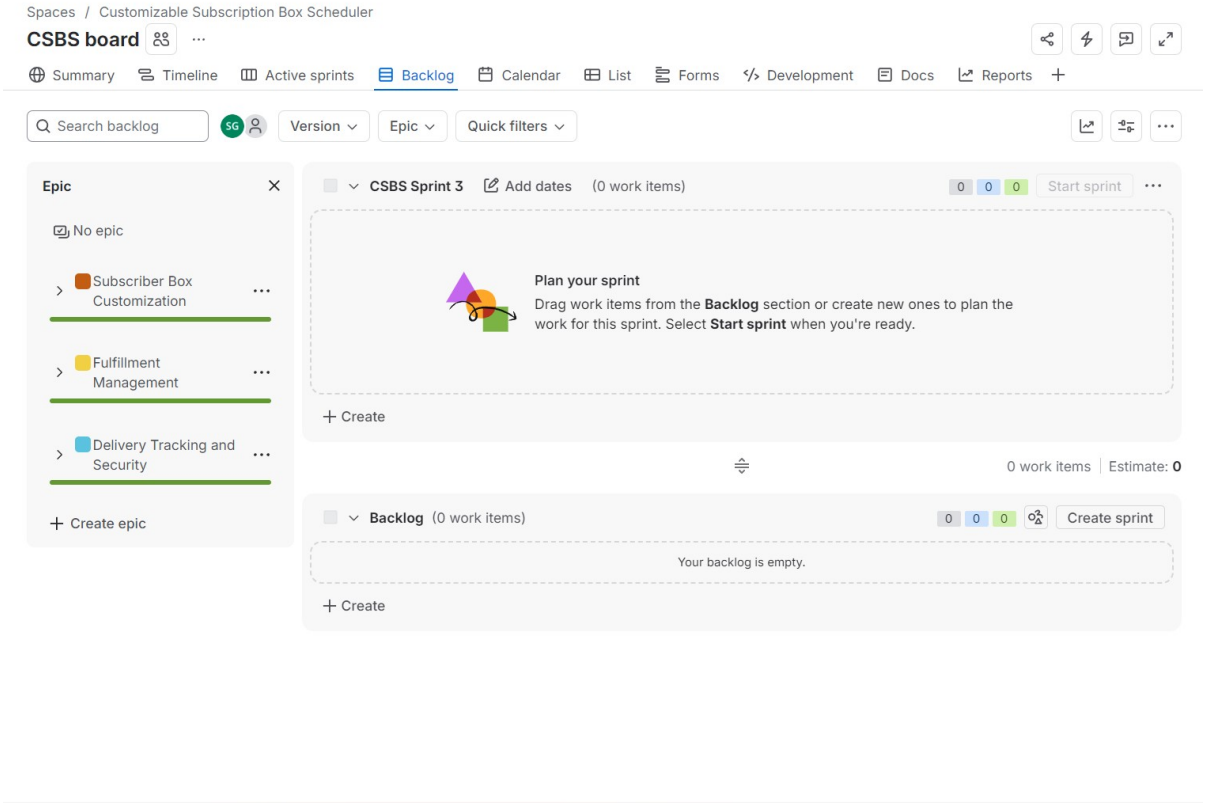
Sprint completion — CSBS Sprint 2

Active sprint view at completion: 0 To Do, 0 In Progress, 1 Done.



Backlog after Sprint 2

CSBS Sprint 3 created automatically with an empty backlog; Sprint 3 was not started.



5. Reflection

1. Did your estimations reflect the actual effort?

Yes. The Fibonacci estimates gave a useful relative measure of complexity, effort, risk, and uncertainty. The smaller stories such as preference filtering and billing-cycle skipping were estimated at 3 points, while more complex operational or performance/security stories received 5 or 8 points. The 37-point Sprint 1 commitment was completed, so the estimates were reasonable for this simulation. Story Points represent relative effort rather than exact hours.

2. Was your backlog well-prioritized?

Yes. High-priority items were assigned to the core subscriber and security/performance requirements because they have direct user impact or important system constraints. The Medium-priority fulfillment-manifest and delivery-status stories were also included to cover the operational flow. This kept the backlog structured around business and user value rather than implementation convenience.

3. How did your simulated sprint align with your plan?

Sprint 1 aligned well with the plan: the seven core User Stories were selected, estimated at 37 points, and progressed through To Do, In Progress, and Done. Sprint 2 was created as the required second sprint, and its single validation story, CSBS-11, was carried through the same workflow — To Do, In Progress, and Done — until Sprint 2 was completed. Both sprints closed with 100% of their committed points, which the burndown charts and the compressed timeline in Section 3 explain.

4. What insights did the Burndown Chart give about your team's capacity?

Taken at face value, both charts fall to zero almost immediately, which would normally flag under-commitment rather than strong velocity. In this simulation that shape is expected, since the whole backlog was moved through the board in one sitting rather than a real week. The exercise still shows what a healthy chart looks like when work is genuinely spread across a sprint: the guideline is the target pace, and a real team's remaining-work line should track close to it rather than dropping instantly.

7. Conclusion

The Lab 1 requirements for the Customizable Subscription Box Scheduler were transformed into three Epics and seven core User Stories in Jira, prioritized and estimated with Fibonacci Story Points for a 37-point Sprint 1. A second, single-story sprint validated that release and satisfied the lab's two-sprint requirement. Both sprints were carried through To Do, In Progress, and Done to full completion, and the resulting Burndown Charts — read honestly rather than at face value — show a compressed classroom simulation rather than a five-day team velocity, which is exactly what this exercise was designed to produce.

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